Ethan McCartney

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Github: https://github.com/Master-Pr0grammer   Website: master-pr0grammer.github.io/personal_website/

Education

Rensselaer Polytechnic Institute (Cumulative GPA: 3.6) — **Graduated Dec. 2025**, ***Troy, NY***

- B.S. Computer Science - GPA: 3.84
- B.S. Mechanical Engineering - GPA: 3.54
- Artificial Intelligence Minor
- Dean's Honor List & Member of National Society of Leadership and Success (NSLS).

Work Experience

Reinforcement Learning Research Assistant — **Jan. 2023 - Jan. 2025**

- Developed and evaluated an RL policy-gradient ('L2A') method for Ising spin glass/MaxCut problems using deterministic REINFORCE, integrating local search and large-scale parallel sampling.
- Matched best-known benchmark solutions on GSet MaxCut and Ising instances, and improved results on larger graphs demonstrating scalability.

Systems Engineer Intern, Potdevin Machine — **May 2023 - Aug. 2023**

- Constructed bill of materials database containing information on raw materials, manufactured parts, routing information, and vendors.
- Automated BOM migration by programming python based automation tools, saving hundreds of manual data entry hours.
- Designed and implemented a new company wide part numbering system.

Physics I / II Tutor & Mentoring — **Aug. 2022 - May. 2023**

- Prepared lessons and conducted two weekly classes of 10-15 students each, reviewing Physics I lectures, practice problems, and quizzes;.
- Proctored several practice exams to help students prepare for exams.
- Provided weekly drop-in individual tutoring sessions with physics I and II students. Reviewed lecture material & homework, covered practice exams, and addressed any academic challenges encountered by students.

Projects

RL Training in Scalable Vectorized GPU Accelerated Environment — **Jan. 2024 - Jul. 2024**

- Vectorized a 2‑D obstacle‑avoidance game in PyTorch, enabling 10k+ parallel simulations on a single consumer GPU.
- Designed a self‑supervised reward signal (positive & soft‑target negative) and trained a neural network over 10 k episodes, increasing average survival time from 4s to 78s, with the final performance rivaling human performance.

Wordle Server from Scratch — **Mar. 2024 - Mar. 2024**

- Built a multithreaded TCP server in C (POSIX sockets) supporting concurrent clients; used mutex-protected shared state and a custom message protocol with robust input validation.
- Implemented clean client lifecycle handling (connect/disconnect/timeouts) and deterministic game logic.

Robotic Arm Kinematics — **Nov. 2023 - Dec. 2023**

- Implemented forward and inverse kinematics for a 6-DOF robotic arm in Python using a Jacobian-inverse approach; integrated PID joint control to stabilize motion and reduce error.
- Implemented real-time object detection/tracking on a uniform background and fused vision feedback with IK to command the arm to follow moving targets.

Language Model from Scratch — **Jul. 2023 - Aug. 2023**

- Implemented a decoder-only Transformer from scratch in PyTorch (causal self-attention, MLP blocks, residuals, layer norm), with full training loop (AdamW, LR scheduling, checkpointing, & early stopping).
- Trained a ~20M parameter character-level model; built preprocessing/data pipeline and evaluation via validation loss/perplexity.
- Implemented beam-search decoding to improve generation quality on small-context models during inference.

C++ Game Engine from Scratch — **Dec. 2023 - Present**

- Built a cross-platform C++ OpenGL engine with model loading, camera/input system, transforms, and basic physics/collision.
- Implemented lighting/shader pipeline (single-light) and real-time rendering; optimized frame loop and debug tools for iteration.
- Achieved over 3,000 FPS with a simplistic scene consisting of a single high fidelity model with lighting running on a RTX 3060ti.

CPU Scheduling Algorithms — **Mar. 2024 - Apr. 2024**

- Implemented and benchmarked FCFS, SJF, SRTF, & Round Robin; measured turnaround time, waiting time, response time, throughput across synthetic workloads and compared tradeoffs.
- Built a workload generator (burst/arrival distributions) and produced a small report summarizing scheduler behavior under I/O-bound vs CPU-bound mixes.

Technical Skills

Languages:	Python, C++, C, Java, MATLAB
ML/Engineering:	PyTorch, Numpy, Simulink
Platforms/Tools:	Linux, MacOS, Windows, Git, Siemens NX (CAD)